

6 Alerting Users via Notifications

6.1 Overview

Android has a whole framework for dealing with pop-up messages, flashing lights, vibration alerts and sound alerts, collectively called notifications, which are handled by the Notification Manager. All Broadcast Receivers, Services and inactive Activities that alert users of events that require attention—for example, incoming calls and appointment reminders—are part of this. A Notification is the only way an application can grab the user's attention if the app is open in the background, and the user isn't paying attention to it.

Applications can add their own status bar icons, with care taken to have them appear only when needed. This can be persisted through insistent repetition. Status bar icons can also be updated regularly or expanded to show additional information using the expanded status bar window.

To raise notifications via the NotificationManager, you need to get the service object via `getSystemService(NOTIFICATION_SERVICE)` from your activity. The NotificationManager gives you three methods: one to pester (`notify()`) and two to stop pestering (`cancel()` and `cancelAll()`). Using the Notification Manager, you can trigger new notifications, modify existing ones or remove those that are no longer needed or wanted.

6.1.1 Hardware Notifications

You can flash LEDs on the device by setting lights to true, also specifying the color (as an #ARGB value in `ledARGB`) and the pattern in which the light should blink (by providing off/on durations in milliseconds for the light via `ledOnMS` and `ledOffMS`).

The following code snippet shows how to turn on the red LED device:

```
notification.ledARGB = Color.RED;
notification.ledOffMS = 0;
notification.ledOnMS = 1;
notification.flags = notification.flags | Notification.FLAG_
SHOW_LIGHTS;
```

Android lets you play any audio file on the phone as a Notification by assigning a location URI to the sound property, as shown in the snippet below:

```
notification.sound = ringURI;
```